[INCH-POUND]
A-A-52524/13
April 10, 1997
SUPERSEDING
MS51001
27 October 1986

COMMERCIAL ITEM DESCRIPTION SPECIFICATION SHEET

SEAL, PLAIN OR SEAL, PLAIN ENCASED: NON-SPRING LOADED, SINGLE SEALING MEMBER

The General Services Administration has authorized the use of this Commercial Item Description Specification Sheet (CIDSS) for all federal agencies.

The complete requirements for procuring the seals shall consist of this document and the issue in effect of A-A-52524.

1. SCOPE. This CIDSS covers seals to be used as rotary shaft seals to retain oil and other fluids.

2. SALIENT CHARACTERISTICS

2.1 Materials.

2.1.1 Class 1 sealing member material. Class 1 sealing members shall be made of polymer-impregnated leather or chrome-vegetable-retanned leather which has been mildew resistant treated and suitable for normal operating temperatures (see A-A-52524).

2.1.2 Class 2 sealing member material. Class 2 sealing members shall be made of oil resistant synthetic elastomer suitable for normal operating temperatures (see A-A-52524).

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any other data which may improve this document should be sent by letter to: U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000.

AMSC N/A FSC 5330

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2.1.3 Class 3 sealing member material. Class 3 sealing members shall be made of oil resistant elastomer suitable for extreme operating temperature service (see A-A-52524).

2.2 Design and construction. The seals (classes 1, 2, and 3 of A-A-52524) shall be designed and constructed in accordance with figure 1.

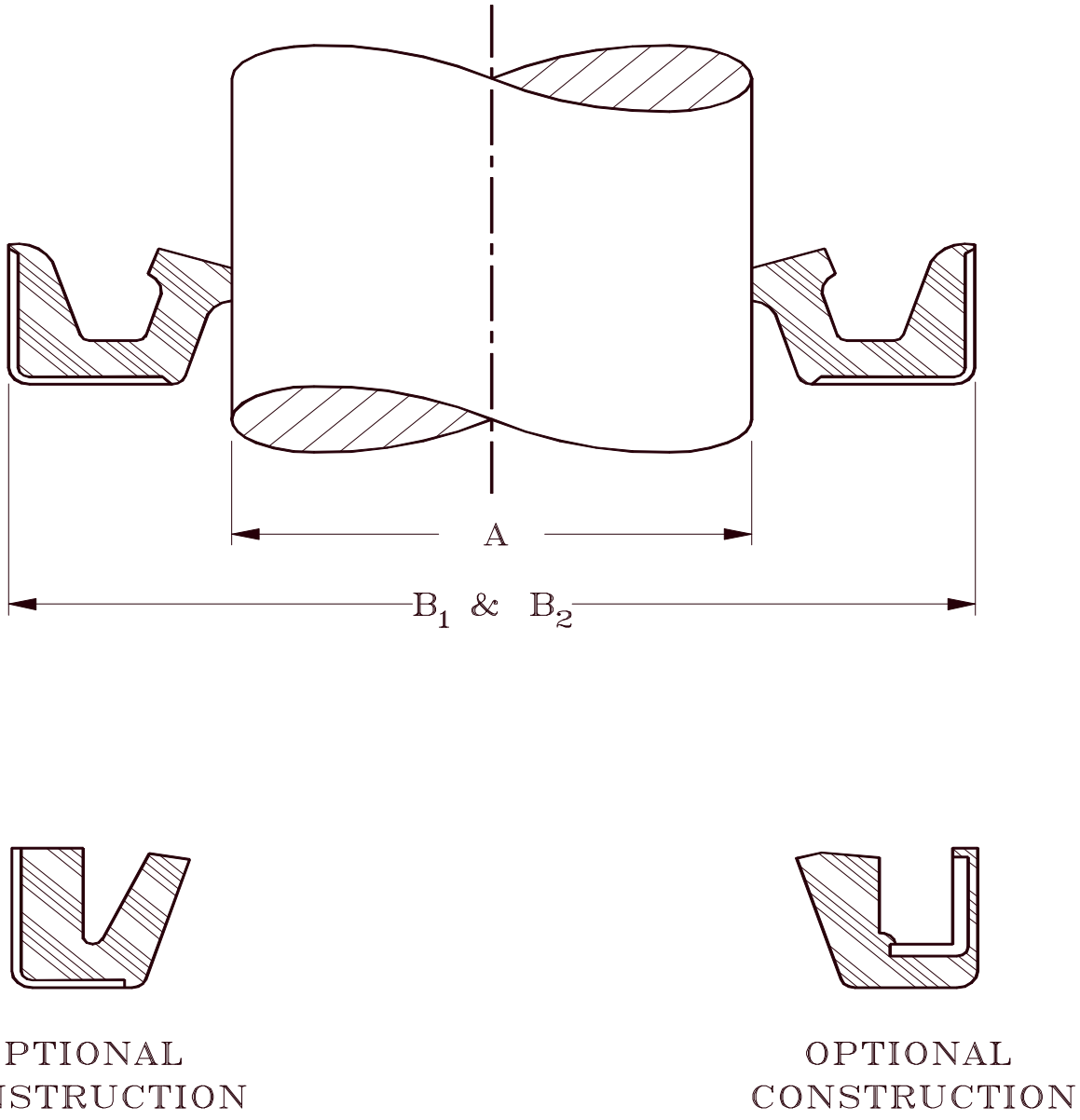

FIGURE 1. Seal design, construction and part number.

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First dash No. (size)	A Shaft dia. (Nom)	B1 seal O.D. (1)	B2 Housing bore (ref)	C Width (5)
1	0.25	0.504	0.500	0.1875
2	0.3125	0.754	0.750	0.25
3	0.375	0.754	0.750	0.25
4	0.4375	1.028	1.024	0.25
5	0.5	1.129	1.125	0.25
6	0.56	1.380	1.375	0.25
7	0.625	1.254	1.250	0.25
8	0.68	1.254	1.250	0.375
9	0.75	1.254	1.250	0.25
10	0.8	1.254	1.250	0.3125
11	0.875	1.254	1.250	0.25
12	0.937	1.380	1.375	0.25
13	1	1.504	1.500	0.25
14	1.0625	1.504	1.500	0.25
15	1.125	1629	1.625	0.25
16	1.1875	1.629	1.625	0.25
17	1.25	1.629	1.625	0.25
18	1.3125	1.754	1.750	0.25
19	1.375	2.066	2.062	0.25
20	1.4375	2.066	2.062	0.25
21	1.5	2.254	2.250	0.25
22	1.56	2.097	2.093	0.25
23	1.625	2.380	2.375	0.25

First dash No. (size)	A Shaft dia. (Nom)	B1 seal O.D. (1)	B2 Housing bore (ref)	C Width (5)
24	1.68	2.380	2.375	0.25
25	1.25	2.380	2.375	0.25
26	1.81	2.380	2.375	0.1875
27	1.875	2.506	2.502	0.25
28	1.94	2.380	2.375	0.1875
29	2	2.380	2.375	0.25
30	2.125	2.504	2.500	0.1325
31	2.187	2.720	2.716	0.25
32	2.25	3.067	3.062	0.4375
33	2.375	2.870	2.875	0.1325
34	2.437	2.941	2.937	0.1875
35	2.5	3.880	3.875	0.25
36	2.25	3.505	3.500	0.25
37	2.87	3.630	3.625	0.4375
38	3	4.130	4.125	0.4375
39	3.125	5.005	5.000	0.25
40	3.375	6.005	5.000	0.25
41	3.5	5.010	5.005	0.25
42	3.625	4.380	4.375	0.25
43	3.75	4.380	4.375	0.375
44	4	5.005	5.000	0.1325
45	4.25	4.880	4.875	0.25
46	4.81	6.005	6.000	0.25

NOTES:

1. Dimensions are in inches. Tolerances are ±0.002 for seal outside diameters (O.D.) up to and including 2 inches. Tolerances are +0.003, -0.002 for seal O.D. over 2 inches and up to and including 3 inches. Tolerances are +0.004, -0.002 for seal O.D. over 3 inches and up to 7 inches.
2. The part number consists of the CID designator, the CID number, the specification sheet number and 2 dash numbers (see paragraph 7.3 of A-A-52524A). The first dash number indicates the size. The second dash number indicates the class.
   EXAMPLE: AA52524/13-11-3 indicates size number 11 with class 3 sealing member (elastomer for extreme operating temperature).

FIGURE 1. Seal design, construction and part number - Continued.

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3. The PIN dash numbers and the former MS part number dash numbers are the same. EXAMPLE: AA52524/13-11-3 cross references to MS51001-11-3.
4. The CIDSS is not intended to limit construction to features other than as shown herein by dimensions, notation, or referenced documents.
5. Width tolerances for seals shall be ± 0.015 for widths up to 0.400 and ± 0.020 for widths over 0.400.

FIGURE 1. Seal design, construction and part number - Continued.

3. NOTES. (This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

3.1 Part or Identification Number (PIN). The PINs to be used for acquisition of seals are shown in paragraph 7.3 of A-A-52524.

3.2 Cross reference data. Seals conforming to this CIDSS are interchangeable/substitutable with seals conforming to MS51001.

MILITARY INTERESTS:

Custodian:
Army - AT

Review Activities:
Army - AR
DLA - IS

CIVIL AGENCY COORDINATING ACTIVITY:
GSA - FSS

Preparing Activity:
Army - AT

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